Surveyor

Surveyor: Decentralized Research, Survey Participation and Reward Distribution on BOT Chain

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Built with HTML5, CSS3, vanilla JavaScript, Solidity and ethers.js on BOT Chain

1. Abstract

Surveyor is a decentralized survey marketplace on BOT Chain. Organizations publish surveys, deposit a reward pool in BOT or USDT, and the pool is shared equally among every respondent whose submission is approved. Surveys, questions, response records, pool balances, reward allocations and payments live in smart contracts, so every participant can verify how a campaign was funded and paid. Rewards are claim-based, which avoids loops over respondent lists and keeps gas costs predictable. This paper describes the problem, the product, the reward formula, the validation and withdrawal systems, the contract architecture and the privacy model.

2. Introduction

Research depends on people answering questions honestly, yet the people who supply those answers rarely see how the reward budget is set or divided. Surveyor puts the budget in escrow at publication and applies one public rule at the end of the campaign. The application is a static web page that talks to a contract through ethers.js; no backend, database or file-storage service is required.

3. Research Industry Challenges

- **Opaque incentives.** Respondents are promised rewards that depend on private spreadsheets and manual payouts.
- **Delayed or missed payment.** Small payouts across many people are costly to send and easy to postpone.
- **Fraud and duplicate responses.** Reward programs attract repeat and low-effort submissions.
- **Weak trust between parties.** Creators cannot prove they funded a campaign, and respondents cannot prove they completed one.
- **Cross-border friction.** Traditional payment rails make small international payouts expensive.

4. Problem Statement

There is no neutral place where a survey creator can lock a reward budget, collect responses, validate them, and distribute payment by a rule that both sides can inspect in advance. Surveyor addresses this with escrowed pools, an explicit approval step, a deterministic equal-split formula and withdrawals that the respondent initiates.

5. Product Vision

Surveyor aims to be the default venue for paid research on BOT Chain: simple enough for a small organization to launch a survey in minutes, and strict enough that a respondent can see before answering how much the campaign holds and how it will be divided. Three roles use the platform: **survey creators** (organizations), **respondents**, and **administrators** who moderate campaigns, resolve appeals, configure fees and minimum withdrawals, and can ban abusive wallets.

6. Survey Creation

Creators register an organization name, then build a survey with a title, deadline, optional response limit and question list. Supported question types are short text, long text, single choice, checkboxes, dropdown, rating (1 to 5), yes/no, number and date. Questions can be marked required and can be shown conditionally based on an earlier answer. Creators can preview a survey and keep a draft in their browser. On publication the question list is stored in the contract as JSON, and every survey receives a unique ID that forms its shareable link. Drafts are local to the browser and are not stored on-chain.

7. Campaign Funding

A campaign is funded when it is created. The creator deposits the reward pool either as native BOT (sent with the transaction) or as USDT (approved and pulled by the contract). The creator may add to the pool while the survey is open. The pool is held by the contract and cannot be moved by the creator until the campaign is finalized. When creating a campaign the creator also chooses a refund policy: return the pool to the creator if no valid submissions exist, or send it to platform fees.

8. Respondent Workflow

- Connect a wallet on BOT Chain and browse surveys or open a public survey link.
- Read the requirements and answer the questions. One submission is allowed per wallet per campaign.
- The submission is recorded with a timestamp and shows as **Submitted**.
- The creator approves it (**Approved**) or rejects it with a reason (**Rejected**). Rejected respondents can appeal.
- After the campaign is closed and finalized, approved respondents claim their reward into their platform balance.
- Once the balance reaches the minimum, the respondent requests a withdrawal.

9. Equal Reward Distribution Formula

Rewards are calculated only after the campaign is closed and finalized. Let *F* be the platform fee in basis points, *P* the pool and *V* the number of valid (approved) submissions, all in integer token units.

```
fee = P x F / 10000
distributable = P - fee
reward per valid respondent = distributable / V (integer division)
remainder = distributable mod V
```

Remainder rule. Approved respondents are numbered 0, 1, 2 ... in the order they were approved. The first *remainder* respondents receive one extra base unit. Because the remainder is always smaller than V, the whole distributable amount is allocated and nothing is left stranded. Example: 10 units shared by 3 valid respondents gives 3 each with a remainder of 1, so the first approved respondent receives 4 and the other two receive 3.

Worked example. A pool of 100 USDT (6 decimals, so 100,000,000 units) with 200 valid submissions:

Fee setting	Fee	Distributable	Per respondent
0 basis points	0	100,000,000 units	500,000 units = 0.5 USDT

Fee setting	Fee	Distributable	Per respondent
250 basis points (default)	2,500,000 units	97,500,000 units	487,500 units = 0.4875 USDT

The 0.5 USDT result assumes no platform fee. The contract default is 2.5%, and the administrator can set any fee up to 10%. If there are no valid submissions, the configured refund policy applies and no fee is taken from a refunded pool.

10. Validation Mechanism

- **Duplicate prevention:** the contract rejects a second submission from the same wallet for the same campaign.
- **Eligibility:** banned wallets and the survey creator cannot submit; the optional response limit stops new submissions when reached.
- **Deadlines:** submissions after the deadline are refused, and anyone can close a campaign once its deadline has passed.
- **Manual approval:** only approved submissions count toward the reward split. Submitted and approved are distinct states.
- **Rejection reasons and appeals:** a rejection stores a reason; the respondent can appeal and an administrator makes the final decision.
- **Finality:** approvals and rejections stop when the campaign is finalized, and no reward is credited before then.

11. Minimum Withdrawal System

Rewards from every campaign accumulate in a per-token balance (BOT and USDT tracked separately). The administrator sets a minimum withdrawal for BOT and for USDT independently. A withdrawal succeeds only if the amount is at least the minimum and not more than the respondent's balance. The balance is reduced before funds are sent, which prevents double withdrawals and re-entrancy. Balances can only change through the contract's own claim, refund and withdrawal logic.

12. Smart Contract Architecture

Surveyor is a single Solidity contract (version 0.8.20 or later) with the following responsibilities:

Area	Functions
Organizations and surveys	registerOrg, createSurvey, fund
Responses	submit, review, appeal, getResponders
Campaign lifecycle	close, finalize (states: Open, Closed, Finalized)
Payouts	claim, withdraw, creator refund via finalize
Administration	setFee, setMin, ban, setAdmin, withdrawFees

Distribution is claim-based: finalization stores two numbers per campaign (the per-respondent amount and the remainder), and each respondent's share is computed in constant time when they claim. No function

iterates over all respondents to pay them.

13. Payment Architecture

Native BOT is represented by the zero address and USDT by the token address supplied at deployment. Pools, platform fees and user balances are tracked per token. Platform fees accumulate in the contract and are collected by the administrator. USDT decimals are read as 6 by the reference front end and should be confirmed for the deployed token.

14. Privacy and Security

Everything stored on-chain is public. Survey questions, response data, wallet addresses and approval status can be read by anyone. Passwords, identity documents and private personal information must never be placed in a survey or an answer.

- **Hash-only responses:** a respondent may store a keccak256 hash of their answers instead of readable text, which commits to the answers without publishing them.
- **Encrypted storage:** creators who need confidentiality can ask respondents to submit client-side encrypted payloads; the reference front end does not yet automate this.
- **Contract safeguards:** checks-effects-interactions ordering on payouts, bounded fee setting, role checks on review and administration, and no unbounded loops in state-changing functions.
- **Audit:** the contract should receive an independent security audit and testnet trial before holding real funds.

15. Business Model

Surveyor charges a configurable platform fee on each finalized campaign that has valid submissions (2.5% by default, capped at 10%). Fees are held per token and withdrawn by the administrator. Pools with no valid submissions and a no-refund policy are also credited to platform fees. Possible future revenue lines include featured placement for surveys and premium organization tooling.

16. Roadmap

Phase	Planned scope
1. Testnet launch	Core contract, survey builder, respondent and creator dashboards, admin controls
2. Hardening	Independent audit, fraud review queues, richer analytics and per-question summaries
3. Privacy	Automated client-side encryption of responses, optional key sharing with creators
4. Growth	QR sharing, organization templates, additional reward tokens, localized interfaces

The roadmap is indicative and may change as the platform and BOT Chain evolve.

17. Conclusion

Surveyor replaces trust in spreadsheets and manual payouts with escrowed pools, visible validation states and a single equal-split formula. Creators get verified participation, respondents get predictable and

claimable rewards, and the whole record can be inspected on the BOT Chain explorer. The remaining work is audit, testing on real campaigns and expanding privacy tooling.

Links: BOT Chain https://botchain.ai • Explorer https://scan.botchain.ai